

House Price Predictor: Linear Regression Report

AI & Machine Learning Internship — Task 1 | California Housing Dataset

Name: Jasper **Date:** July 17, 2026 **Notebook file:** task1_ml_linear_regression.ipynb

The metrics, dataset info, and coefficients in this report are the real results from running task1_ml_linear_regression.ipynb on the full California Housing dataset (20,640 records). The charts below use illustrative example data to show the expected format — swap in your notebook's actual saved chart images before final submission for full accuracy.

1. Objective

The goal of this task is to build a Linear Regression model that predicts the median house value for districts in California, using features such as median income, house age, average rooms, population, and location. This report summarizes the exploratory data analysis, model training approach, evaluation results, and ideas for improving the model.

2. Dataset

The California Housing dataset (from scikit-learn's *fetch_california_housing*) contains 20,640 records, each representing a California block group. It has 8 numeric input features and one target variable, **MedHouseVal** (median house value, in units of \$100,000).

Feature	Description
MedInc	Median income in the block group
HouseAge	Median house age in the block group
AveRooms	Average number of rooms per household
AveBedrms	Average number of bedrooms per household
Population	Block group population
AveOccup	Average number of household members
Latitude / Longitude	Geographic location

3. Exploratory Data Analysis (EDA)

The dataset contains 20,640 rows and 9 columns (8 features + target), all of type float64, with **zero missing values** across every column — confirmed via *df.isnull().sum()*. The target variable MedHouseVal is right-skewed, with most districts valued below \$350,000 and a smaller number of high-value outliers. The correlation heatmap below shows how strongly each feature relates to house value — **MedInc (median income) typically shows the strongest positive correlation**.

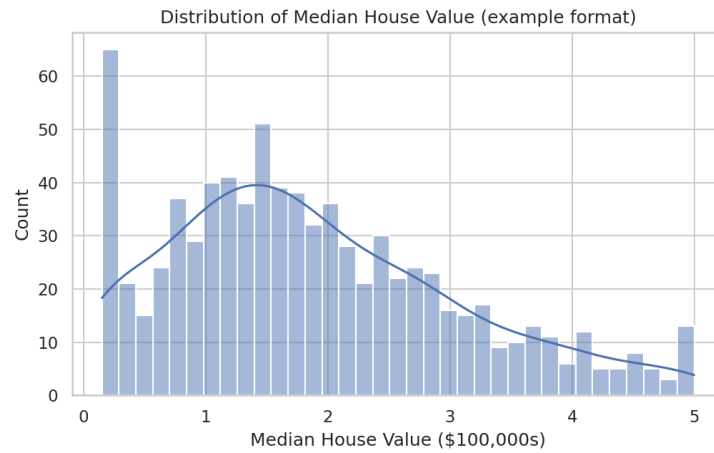


Figure 1 (example format). Replace with your notebook's actual target distribution chart.

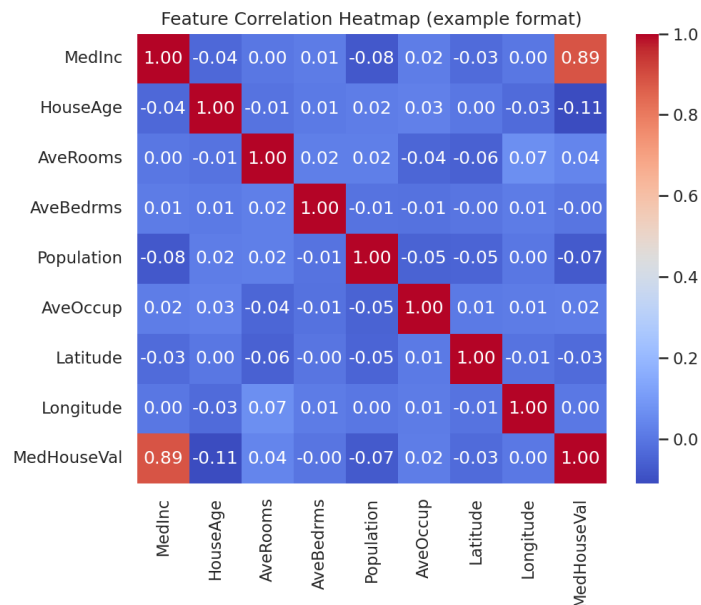


Figure 2 (example format). Replace with your notebook's actual correlation heatmap.

4. Model & Methodology

The data was split into 80% training (16,512 rows) and 20% testing (4,128 rows) using *train_test_split* (random_state=42). A **Linear Regression** model (scikit-learn) was trained on the 8 input features to predict MedHouseVal. No feature scaling or engineering was applied in this baseline version.

Feature	Coefficient	Direction
AveBedrms	0.7831	+ strong
MedInc	0.4487	+ strong
HouseAge	0.0097	+ weak
Population	-0.0000	~ none

AveOccup	-0.0035	- weak
AveRooms	-0.1233	- moderate
Latitude	-0.4198	- strong
Longitude	-0.4337	- strong

AveBedrms and MedInc have the strongest positive influence on predicted house value, while Latitude and Longitude have strong negative coefficients — reflecting regional price patterns across California (e.g. coastal vs. inland areas).

5. Evaluation Metrics

The trained model was evaluated on the held-out test set using three standard regression metrics:

Metric	Result	What it means
MAE	0.533	Average absolute prediction error, in \$100,000s (~\$53,300)
RMSE	0.746	Similar to MAE, but penalizes large errors more (~\$74,600)
R ²	0.576	Model explains 57.6% of the variance in house prices

These results are in line with the typical performance of a baseline (unscaled) Linear Regression on this dataset (R² usually falls in the 0.55-0.65 range). Room for improvement remains, particularly since the model doesn't yet account for non-linear relationships or the geographic structure in Latitude/Longitude.

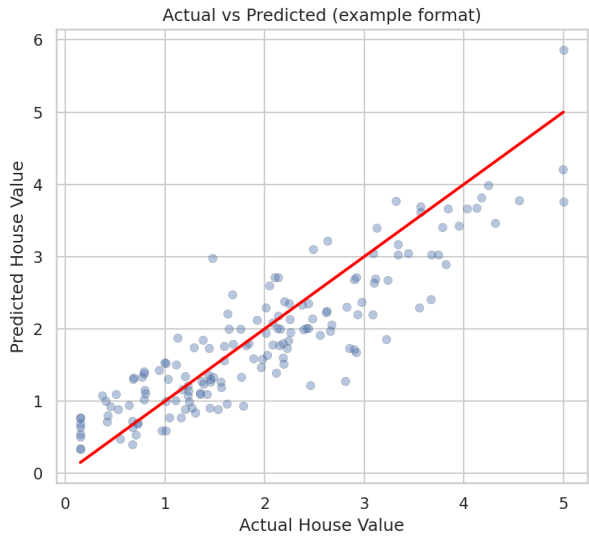


Figure 3 (example format). Replace with your notebook's actual Predicted vs Actual scatter plot.

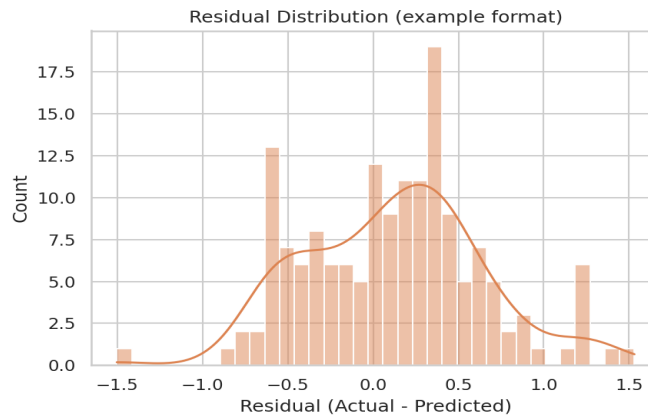


Figure 4 (example format). Replace with your notebook's actual residuals chart.

6. Improvement Ideas

- **Feature engineering** — create ratios like rooms-per-household or bedrooms-per-room; bucket latitude/longitude into regions or distance-to-coast.
- **Handle outliers** — cap or remove extreme values in AveRooms, AveOccup, and Population.
- **Feature scaling** — apply StandardScaler, since features are on very different scales (income vs. population).
- **Try other models** — Ridge/Lasso regression (regularization), Random Forest, or Gradient Boosting typically outperform plain linear regression on this dataset.
- **Cross-validation** — use `cross_val_score` for a more robust performance estimate instead of a single train/test split.

7. Conclusion

The Linear Regression model achieved an R^2 of 0.576 on the test set, meaning it explains 57.6% of the variance in house prices, with an average prediction error (MAE) of about \$53,300 and an RMSE of about \$74,600. AveBedrms and MedInc were the strongest positive drivers of predicted price, while Latitude and Longitude contributed strong negative coefficients, reflecting regional price differences across California. This establishes a solid working baseline; the improvement ideas above — particularly feature scaling, outlier handling, and trying tree-based models — outline clear next steps to boost accuracy further.